

Mini quiz 1

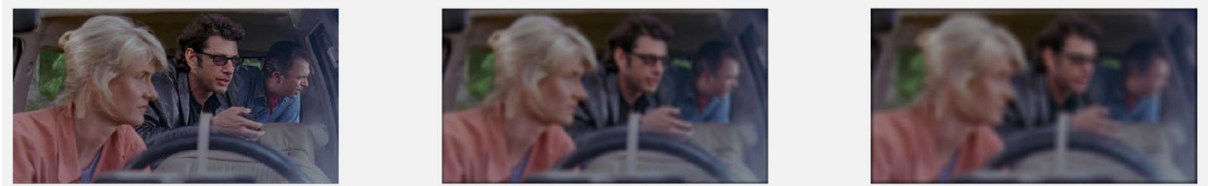
Which of the following statements are **TRUE**?

(you can select more than one, if necessary)

- ☐ Harris corner detector cannot estimate the orientation of a feature
- ☒ SIFT features are invariant in scale differences
- ☒ Lines are not good features to track
- ☒ The SIFT feature descriptor has 64 values
- ☐ Kd-Tree performs half of the operations compared to a greedy search.

Question 2

0 / 1 point



In the attached graphic, the left-most image is the original image. To create the middle image, the original was treated with a 9x9 mean filter.

What filter could have been used to produce the right-most image from the original (left-most) image?

(Choose all answers that might apply)

- ☐ 9x9 median filter
- ☒ 5x5 median filter
- ☒ 15x15 mean filter

Question 3

1 / 1 point

Consider a grayscale digital image with 100 rows and 460 columns, where each pixel can take values from 0 up to 1023.

How many bits are required to store such a digitized image?

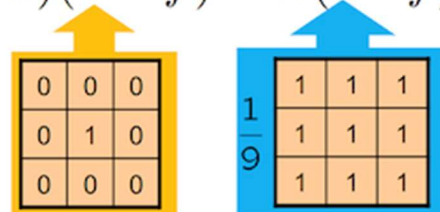
Answer:

460,000

Question 4

1 / 1 point

$$\begin{aligned}
 f_{sharp} &= f + \alpha(f - f_{blur}) \\
 &= (1 + \alpha)f - \alpha f_{blur} \\
 &= (1 + \alpha)(w * f) - \alpha(v * f)
 \end{aligned}$$



$$= ((1 + \alpha)w - \alpha v) * f$$

We have seen in our lectures that we can implement a sharpening linear filter by a single convolution with an appropriate kernel, as shown in the attached graphic. Assume that $\alpha=0.9$ to calculate the sharpening filter kernel.

Which of the following tables is the sharpening kernel in this case:

- ☐

-0.9	-0.9	-0.9
-0.9	1	-0.9
-0.9	-0.9	-0.9
- ☐

0	0	0
0	-1	0
0	0	0
- ☒

-0.1	-0.1	-0.1
-0.1	1.8	-0.1
-0.1	-0.1	-0.1

Question 5

0 / 1 point

Which of the following kernels calculates the horizontal gradient?

- ☐

-1	0	1
-2	0	2
-1	0	1
- ☒

0	0	0
---	---	---

- ☐

-1	0	1
0	0	0
- ☐

0	-1	1
0	-1	1
0	-1	1
- ☒

0	0	0
0	-1	1
0	0	0
- ☐ None of the above

mini-Quiz 2

Consider the following two tables and calculate their dissimilarity using as metric the sum of squared differences.

$$A = \begin{pmatrix} 10 & 15 & 20 \\ 20 & 20 & 25 \\ 10 & 15 & 20 \end{pmatrix} \quad B = \begin{pmatrix} 15 & 15 & 15 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{pmatrix}$$

Answer: ✗ (800)

Question 2 0 / 1 point

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

Assuming this camera matrix please fill in the correct correspondences:

- ✗ _____ (2) f
- ✗ _____ (4) c
- ✗ _____ (3) e
- ✗ _____ (5) i
- ✗ _____ (1) a

- 1. fx
- 2. cy
- 3. fy
- 4. cx
- 5. 1

Question 3

Consider a stereo vision system.

Choose all the statements below that are true.

- ➡✗ ☐ The epipoles lie on the baseline-containing line
- ➡✗ ☐ All epipolar lines intersect at the epipoles
- ➡✗ ☐ The epipoles can be outside the images
- ☐ All epipolar lines meet at the optical center
- ☐ All epipolar lines are parallel to the optical axis
- ☐ The baseline intersects the epipolar plane at the epipoles

Question 4

0 / 1 point

What does a low disparity value mean?

- ☐ The matching pixels are found far from the same position in both images
- ➡☐ The matching pixels are found near the same position in both images
- ☐ Disparity does not tell us about the position of the pixels, but about how similar they are
- ☐ There are no matching pixels between the two images

Question 5

0 / 1 point

Order the following stereo vision algorithms, according to their typical computational complexity:
(where 1 is LEAST computational demanding, and 3 is MOST computational demanding)

- ✗ _____ (2) Dense Dynamic Programming-based Stereo Vision Algorithm
- ✗ _____ (3) Dense Graph Cuts-based Stereo Vision Algorithm
- ✗ _____ (1) Dense Local Stereo Vision Algorithm

Question 6

Camera Projection

Assuming a camera at location $[X \ Y \ Z] = [0, 0, 0]$ with a pose defined by the following rotation matrix:

[1, 0, 0]
[0, 1, 0]
[0, 0, 1]

given a camera matrix

[725, 0, 631]
[0, 726, 360]
[0, 0, 1]

At which "x **camera coordinate**" would the following 3D point be depicted:

[X, Y, Z] = [1,1,5]

Answer:

✖ (776)

Question 70 / 1 point

Choose all the statements below that are true.

- ☐ The fundamental matrix projects a 3D point in the right camera frame to a 2D point in the left image frame.
 - ☐ The fundamental matrix projects a 3D point in the right camera frame to a 3D point in the left camera frame.
 - ☒ The essential matrix includes the pose of the cameras with respect to each other.
 - ☒ The fundamental matrix projects a point in the right image frame to a point in the left image.
-

mini-Quiz 3

Which of the following algorithms requires as input explicit matches between point pairs from two point clouds BEFORE it starts executing?

- ☐ FPFH
- ☐ Spin Images
- ☐ ICP
- ☒ Kabsch algorithm

Question 2

0 / 1 point

Please select the correct among the following statements concerning the Iterative Closest Points (ICP) algorithm:

- ☐ ICP is guaranteed to provide the optimal transformation that best aligns 2 point clouds
- ☒ ICP provides a rigid transformation between 2 point clouds
- ☒ ICP works best if an initial rough estimation of the alignment of 2 point clouds is known.
- ☐ ICP works best if the 2 point clouds are far from each other.
- ☐ ICP provides an affine transformation between 2 point clouds

Question 3

0 / 1 point

Which of the following clustering results could have been produced using k-means?

(choose all correct answers)



A:



B:



C:



D:



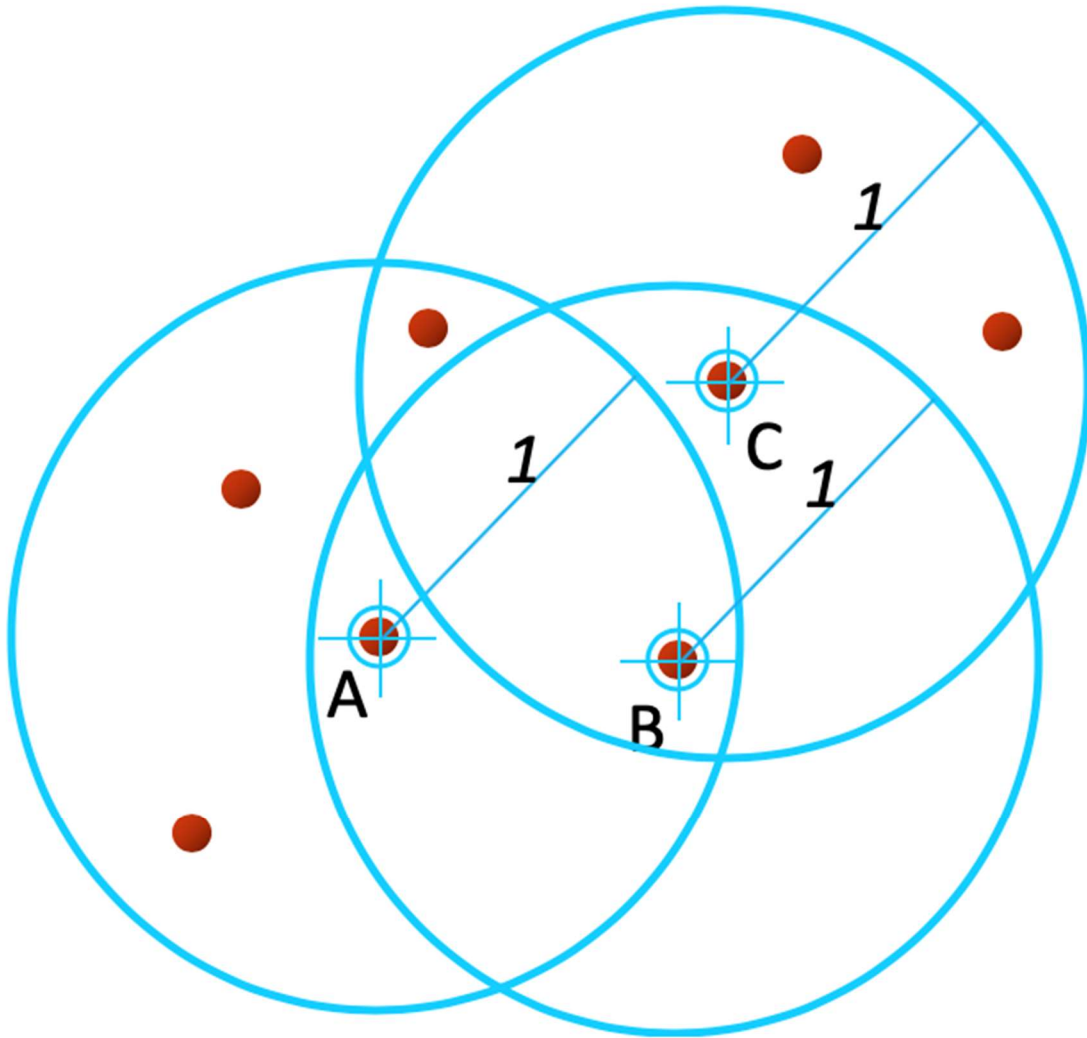
E:

- ☐ A
- ☐ B
- ☒ C
- ☐ D
- ☒ E

Question 4

Consider the following points.

0 / 1 point



How would points A, B and C be characterized according to DBSCAN?

(assume $\epsilon=1$ and $MinPts=4$)

- ☐ A: core | B: core | C: core
- ☐ A: noise | B: core | C: noise
- ☐ A: core | B: border | C: border
- ☐ A: core | B: noise | C: core
- ☐ A: border | B: noise | C: border
- ☐ A: noise | B: border | C: noise
- ☒ A: core | B: border | C: core
- ☐ A: noise | B: border | C: core

Question 5

0 / 1 point

Consider a color image with dimensions 640x480 pixels. We would like to cluster its pixels making full use of both geometric and color information (including intensity/lightness).

How many clusters will the Mean Shift algorithm consider initially? (provide the exact number below)

Answer:  (307200, 307,200, 307.200)

Question 6 0 / 1 point

Consider a color image with dimensions 640x480 pixels. We would like to cluster its pixels making full use of both geometric and color information (including intensity/lightness).

What would be the dimensionality of the problem in this case?

- ☐ 4
- ☐ 2
- ☒ 5
- ☐ 3

Question 7 0 / 1 point

Given the dataset 'clusters.txt' ([link](#)) that has been read into the variable X, find how many clusters are optimal for this dataset using the elbow method.

Recall you call Kmeans:

```
km = KMeans(n_clusters = n)
```

```
km.fit(X)
```

- ☐ 2
- ☐ 3
- ☐ 4
- ☒ 5
- ☐ 6
- ☐ 7

Question 8 0 / 1 point

Apply Linear Regression to the provided data ([link1](#), [link2](#)) to obtain a model of the form $y=ax+b$.

Follow the provided guidelines for the implementation.

What are the values of the parameters "{a}" and "{b}" in that model (APPROXIMATELY)?

- ☐ $a = 0.23, b = 25.60$
- ☒ $a = 0.55, b = 19.86$
- ☐ $a = 0.82, b = 19.86$
- ☐ $a = 0.55, b = 12.37$
- ☐ $a = 0.23, b = 12.37$
- ☐ $a = 0.82, b = 25.60$

mini-Quiz 4

Which of the statements below are correct for PCA?
(choose all that apply)

- ☒ ☒ PCA relies on the calculation of eigenvectors and eigenvalues
- ☐ PCA dictates how many dimensions need to be retained
- ☒ ☒ PCA is a dimensionality reduction technique
- ☒ ☒ PCA in a 5-dimensional space can create exactly 5 new features.
- ☐ the PCA output describes the significance of each of the original features

Question 2

0 / 1 point

A SVM draws lines, planes or hyper-planes in the features' space. As a result, it can only treat problems that are linearly separable in the original space.

- ☐ True
- ☒ False

Question 3

0 / 1 point

Consider a classification problem where there are 2 classes. Class "A" contains 26 unique instances, while class "B" contains 5 unique instances.

What is a suitable value for "k" of a k-NN classifier that can correctly classify new instances?

- ☐ $k=5$
- ☐ Any value of k between 5 and 26 (including 5 and 26) is suitable.
- ☒ We don't have enough information to define k
- ☐ Any value of k equal to, or greater than 5 is suitable.
- ☐ $k=2$

Question 4

0 / 1 point

Please rate the following approaches to Relative Pose Estimation in terms of expected accuracy.

(1 being best and 3 worst)

✗ ____ (2) 2

1. 3D to 3D

✗ ____ (3) 1

2. 3D to 2D

✗ ____ (1) 3

3. 2D to 2D

Question 5

Concerning Relative Pose Estimation, please match the problem to the solution

✗ ____ (2) 3D to 2D Relative Pose Estimation

1. ICP

✗ ____ (1) 3D to 3D Relative Pose Estimation

2. PnP

✗ ____ (3) 2D to 2D Relative Pose Estimation

3. Essential Matrix

Question 6

Assume that you just run a PnP solver with:

- some 3D points in World Coordinates and,
- The same points in the Image Coordinate of your current camera pose.

The result was:

rvec = [-0.05, -1.51, -0.00]

tvec = [87.39, -2.25, -24.89]

Assuming that you perform stereo reconstruction on the camera you get an **interest point P** in location $X,Y,Z = [-6.71, 0.23, 21.59]$ in camera coordinates.

What would be the the location of the **interest point P** in world coordinates?

- ☐ 96.75, -1.98, 40.71
- ☐ 96.75, 40.71, -1.98
- ☐ -1.98, 96.75, 40.71
- ☒ 40.71, -1.98, 96.75